ENHANCING SANDHILL CRANE MONITORING WITH DEEP LEARNING AND UNCERTAINTY QUANTIFICATIONS

CRP 2024 PROJECT PROPOSAL

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ABSTRACT

In the recent decade, Deep Learning (DL) methodologies have shown wide range of applications in many fields due to its ability to leverage large raw data and make accurate predictions. Object detection, being one of DL computer vision technique, plays a pivotal role in reducing labor and cost for population monitoring. An accurate index of the population of interest allows for critical decision-making to be deployed in the real-world. Hence, many studies on crowd and wildlife counting have utilized DL to develop an automated counting framework to avoid excessive manual work and observer biases. However, due to the nature of deep neural networks, Uncertainty Quantification (UQ) is a challenge for building reliable AI. Moreover, with the difficulty of collecting sufficient training data in annual surveys, it is of high interest to produce a robust model that is able to generalize into future surveys. In this study, we aim (1) test the scalability of the model and improve its overall accuracy and (2) study the existing methods for UQ applied to a comprehensive data set of MCP sandhill cranes (*Antigone canadensis*) and fine-tune the model to inform generalizability.

1 Introduction

Wildlife population monitoring plays a critical role in conservation ecology, providing essential data to guide efforts in preserving biodiversity and maintaining healthy ecosystems. Many wildlife data collection efforts now rely on remote-sensing technologies, enabling researchers to gather comprehensive ecological information from difficult-to-access environments without direct human presence. Various methods of remote-sensing data collection include the use of thermal imagery, LiDAR, multispectral and hyperspectral imaging, each offering unique insights into environmental and ecological parameters from a distance. These methods usually result in large volume of raw data, which requires careful processing prior conducting any analysis. Hence, it would be costly and time inefficient to analyze the large raw data manually.

Recent advancements in computing hardware, particularly in graphics processing unit (GPU), have enabled a practical use of deep neural networks applied to computer vision problems [Kaur and Singh, 2023]. Object detection, being one of the most challenging domains of computer vision, aims to train a machine to identify and classify objects within images or videos. In object detection tasks, annotations are typically made through bounding boxes, which are drawn around objects in images to provide a visual and spatial reference for machine learning algorithms as illustrated in Figure 1. Therefore, to count objects of interest, one simply needs to count the number of bounding boxes that the model predicts for each class of objects.

Numerous studies have already employed object detection algorithms in wildlife conservation ecology, demonstrating the practical applications and effectiveness of these techniques in diverse ecological studies [John et al., 2024, Hentati-Sundberg et al., 2023, Mitterwallner et al., 2023]. For our particular problem, [Luz-Ricca et al., 2023] applied two DL

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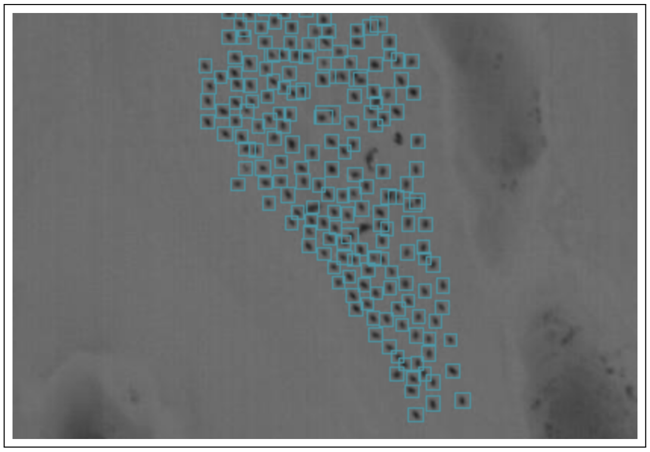


Figure 1: Bounding Box Predictions of Object Detection Model

methods (ASPDNet and Faster R-CNN) on a small batch of data to demonstrate the ability for DL models applied in counting sandhill cranes.

To enhance the efficacy of the model, and to assess the potential to be applied in the wild, we aim to develop a one-stage multi-class object detector with higher overall accuracy and acceptable error. Defining an acceptable error comes down to quantifying any uncertainties, which will be discussed further in the methods section. We will primarily be experimenting with You-Only-Look-Once (YOLO) models: a state-of-the-art computer vision framework.

In conservation, where the stakes are high and resources often limited, being able to measure and communicate the degree of uncertainty in population estimates enables better risk management and decision-making. Uncertainty quantification helps identify where more research is needed to refine predictions, thus guiding efficient allocation of efforts and funding. It also fosters transparency and confidence among stakeholders by clearly outlining the limitations and confidence levels of the findings presented. [Langford et al., 2009] addresses the importance of quantifying uncertainty in any conservation planning, and without it comes with consequences of inaccurate predictions. In our particular problem, uncertainties may arise from model falsely classifying different species, under/over counting objects, predicting backgrounds as objects, inconsistent model predictions, and more.

Modeling uncertainty is crucial for reliable AI system; however, it is more complex to achieve in DL scenarios due to the large number of tuning parameters. As a result, many studies have only provided point estimates of the prediction with no consideration of the underlying uncertainties. Hence, in this work we explore two main branches of UQ applied to DL scenarios: Monte Carlo (MC) Dropout and ensemble modeling. Both methods have their advantages and disadvantages, which will be discussed later in the proposal.

The remainder of this proposal is organized as follows. A literature review of existing methodology will inform the current limitations. Next, each of the research objectives will be introduced followed by their corresponding methods. Finally, the remaining work during Summer 2024 will be outlined with a prospective timeline to shape the final deliverables of the project.

2 Literature Review

2.1 Previous Study of DL Sandhill Cranes

[Luz-Ricca et al., 2023] have demonstrated the feasibility of DL models applied to counting sandhill cranes. The two primary models applied were a detection model Faster R-CNN [Ren et al., 2016] and a density estimation model ASPDNet [Gao et al., 2020].

Faster R-CNN is the two-stage object detector with RPN (Region Proposal Network) and a detection network. It produces candidate bounding boxes through the RPN, which is used in making predictions using the detection head. Faster R-CNN was seminal in many of the later developments in object detection and set a bar for both performance and efficiency. On the other hand, ASPDNet is a recent development in density estimation that trains to match a kernel density estimate derived from the point annotations in imagery, to quantify objects [Lempitsky and Zisserman, 2010]. It starts with producing maps of density, for which each pixel value represents the possibility of object presence at this place. In the usual application, the process of density estimation starts from a Convolutional Neural Network (CNN) used for learning features from training images. Annotations on those images contains not only bounding boxes but also single-point annotations, represented by density maps with the distribution of objects around the images. Hence, CNN is trained to predict these density maps from the input images. It is particularly useful for handling the large number of objects of varying sizes, ambiguity present in overlap, and the tendency of clustering in crowded scenes. Figure 2 illustrates the discrepancy in results of the two methods discussed.

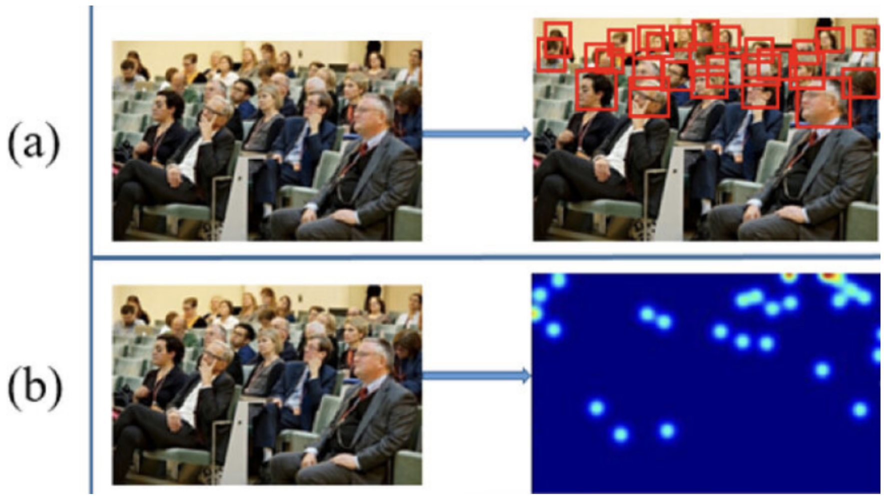


Figure 2: Different Results of (a) Object Detection vs. (b) Density Estimation

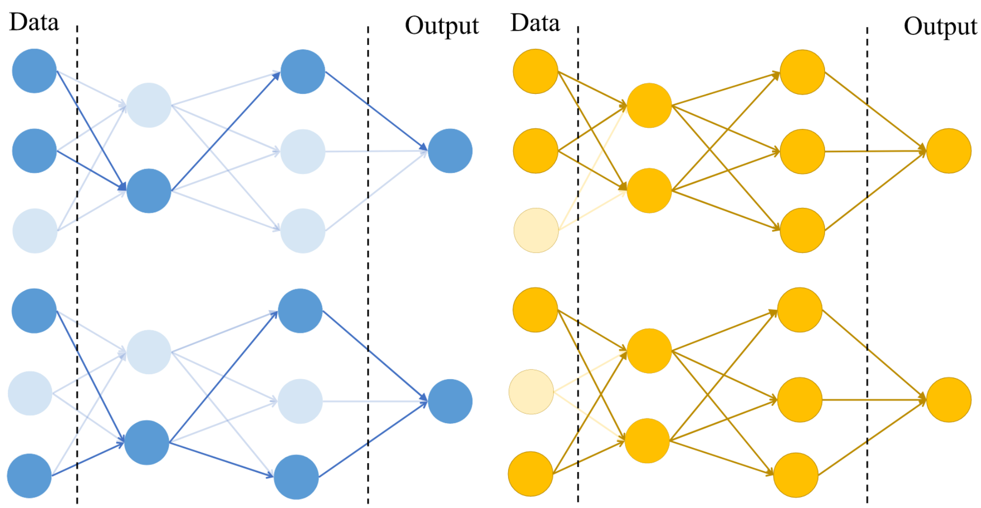
In his findings, ASPDNet outperformed Faster R-CNN in terms of model robustness and general accuracy whereas the Faster R-CNN severely suffered from under counting the object of interest. Despite the satisfactory results of the density estimation method, ASPDNet lacks the ability to spatially locate the objects. Also, depending on the training configurations, ASPDNet have shown inconsistent results when predicting unseen images. Moreover, the results can be unintuitive as it simply outputs pixel values representing the likelihood of an object's presence at that location.

The spatial location of the sandhill cranes is important for making decisions to preserving or modifying the vegetations of the area in highly populated roosting sites. Thus, for practitioners interested in the exact location of sandhill cranes ASPDNet may not be a suitable model. Faster R-CNN, on the other hand, is a detection method that outputs bounding boxes for each object found. Thus, to locate the object of interest, we simply take the midpoint of the bounding box. Hence, the results are more intuitive providing spatial information of the objects. Therefore, in this study we follow the path of detection method with the aim of reaching the satisfactory performance at least of the density estimation method.

2.2 High-level Overview of UQ Methods

In deep learning context, UQ can be classified largely into two categories: Bayesian approximation and ensemble learning techniques [Abdar et al., 2021]. In this work, we aim to apply two UQ methods: Monte Carlo (MC) Dropout and ensemble modeling. Monte Carlo (MC) dropout is a practical and widely-used technique for uncertainty quantification in deep learning models, particularly neural networks. Proposed by [Gal and Ghahramani, 2016] as a Bayesian approximation, MC dropout addresses the inherent overconfidence in predictive models that do not account for uncertainty. By applying dropout not only during training but also during inference, MC dropout simulates the effect of sampling from the posterior distribution of a Bayesian network. During inference, predictions are made multiple times with dropout enabled, leading to a distribution of outputs from which uncertainty can be quantified. This process approximates Bayesian inference without the computational cost associated with full Bayesian methods, making it particularly useful in deep learning applications where model confidence and decision-making under uncertainty are crucial.

Ensemble model works by training multiple instances of the same model but with differences in either initialization, training subsets, or even model architectures and finally using aggregation of their predictions. This diversity among the models gives a richer view of the possible outcomes and, hence, better quantifies the uncertainty. The main advantage of ensemble methods with CNNs for object detection lies in its potential to capture different aspects of data; hence, it brings the variance down while at the same time improving upon accuracy. Ensemble methods, such as bagging and/or boosting, could be adapted to train several detectors, whose varied predictions are then effectively combined and reduced to yield a final decision for quantifying uncertainty and improve detection performance. Other model validation approaches using stratified cross-validation with non-probabilistic methods will continue to be explored.



(a) Monte Carlo (MC) dropout

(b) Bootstrap model

Figure 3: Architecture Comparison of MC Dropout and Bootstrap model

3 Research Objectives / Methods

3.1 Achieving Higher Accuracy

Currently, we have a small batch of dataset annotated from 2018 and 2021 and a new 2023 dataset with large pre-annotations available. The main goal of this project is to get an accuracy as high as possible to capture the true population of MCP sandhill cranes on the new 2023 data set. To achieve this, we will be exploring the state-of-the-art object detection models such as YOLO (You-Only-Look-Once) and RT-DETR (Real-Time Detector) frameworks developed by Ultralytics. Model training and evaluation will be done on both 2018/2021 and 2023 dataset to study the dataset discrepancy and model comparisons.

YOLO models are a family of object detection algorithms that excel in real-time processing by predicting both bounding boxes and class probabilities with a single forward pass through the network. YOLO models divide an image into a grid and predict bounding boxes and probabilities for each grid cell. This approach contrasts with methods that propose regions and then classify them, making YOLO exceptionally fast and suitable for applications requiring real-time processing. Over time, several versions have been released, such as YOLOv1 through YOLOv8, each improving in accuracy, speed, and the ability to detect small objects. [Ammar et al., 2021] compared the performance of Faster R-CNN and various YOLO models and yielded markedly better performance for YOLO models in most configurations on aerial images for car detection. For the best accuracy, we will primarily be experimenting on the latest YOLOv5 and YOLOv8 models with varying configurations.

Training the model on the new 2023 dataset requires training dataset with manual annotation. The process resembles an active learning framework where one trains a model on a small set of manually annotated images, using the model to make further predictions, manually correcting any mispredictions, and then feeding these corrected annotations back into the training dataset to iteratively improve the model's accuracy and performance. For manual annotations, we will be using an open-source data labeling tool Label Studio.

To evaluate the model, the following metrics will be used: mAP50-95 (Mean Average Precision), Mean Percentage Error (MPE), Mean Absolute Error (MAE), and Mean Squared Error (MSE). mAP50-95 measures the model's accuracy by calculating the average precision (AP) at different IoU thresholds from 0.5 to 0.95 (inclusive), in steps of 0.05. This will be a primary metric to determine the goodness of model fit. MPE, MAE, and MSE will be supplementary metrics to evaluate model predictions on subspecific images. Overall, we aim to achieve an MPE < 10%. The following equations will be used to compute the metrics.

$$\text{mAP@}[.50:.95] = \frac{1}{10}\sum_{t=0.5}^{0.95} \text{AP}_t \tag{1}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{2}$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{3}$$

$$\text{MPE} = \frac{100\%}{n}\sum_{i=1}^{n} \left(\frac{y_i - \hat{y}_i}{y_i}\right) \tag{4}$$

where y_i and $\hat{y}_i$ denote the true number of objects and predicted objects, respectively.

Post-processing techniques are oftentimes used to better the prediction by further reducing the prediction errors of the network [Salvi et al., 2021]. It is assumed that most of the prediction errors occur due to the nature of our data set: tiny objects in high resolution image. Thus, it would be helpful to make predictiosn on zoomed-in images by tiling a large image into a specific height-width ratio. Slicing Aided Hyper Inference (SAHI) [Akyon et al., 2022] is a computer vision tool designed to facilitate the detection of both small and large objects in high-resolution images by performing efficient slicing and merging operations during model inference. With its easy integration into many object detection framework, we will adopt this post-processing technique into the YOLO model.

The final predicted count will be performed on a comprehensive dataset of approximately 40,000 1280x720 RGB images. Orthomosaicking is a process in remote sensing that involves stitching together multiple aerial or satellite photographs into a single, continuous image called an orthomosaic. Since these aerial thermal images have approximately 20% overlap, we will be performing predictions on orthomosaicked images to avoid duplicate countings.

To handle the computational burden, we will leverage the A100 GPU in Google Colab as well as the High-Performance Computing (HPC) at William and Mary. The full inference is expected to take <24 hours.

3.2 Differentiating Non-Target Species Analytically

The new dataset acquired in 2023 introduces non-target species, specifically ducks and geese, in the thermal imageries. Hence, an effective approach is needed to enable model to classify different species with an acceptable error. Once the model is fitted, we will conduct an error analysis to identify scenes with the most error (i.e. false positives and false negatives).

Modeling the machine to classify objects into different classes isn't difficult. The problem arises in its ability to accurately detect the different species. Based on a naive analysis of the images, ducks/geese seem to have smaller body size as opposed to sandhill cranes. Furthermore, ducks/geese tend to clutter together, which makes detection harder due to the lack of space between objects. On the other hand, sandhill cranes tend to have uniform spacing within and appears to have white dots around the head due to the nature of their cold head.

Once the model is fitted and used for inference, we will have a better understanding of the model's ability to classify and detect across different species. Conducting an error analysis on the occurrences of false-positives and false-negatives will provide us with an approach further needed to take to accomodate the errors.

Figure 3 depicts a typical image of where target and non-target species coincide together. The larger body objects on the top half are sandhill cranes whereas the smaller dense objects in the bottom half are ducks/geese. As it is difficult for human eyes to differentiate, it would take considerable effort to train a machine capable to classify them.

3.3 Uncertainty Quantification and Prediction Bounds

Towards the end of the modeling, we will deploy two uncertainty quantification methods described above. These will provide us with model uncertainty (epistemic uncertainty) and confidence bounds of model predictions. Through this, we can assure that the model is indeed consistent and robust on varying conditions.

Monte-Carlo Dropout (MC-Dropout) method will allow us to resemble a Bayesian Neural Network without increasing computational complexities [Gal and Ghahramani, 2016]. By providing a probabilistic interpretation of the network, we

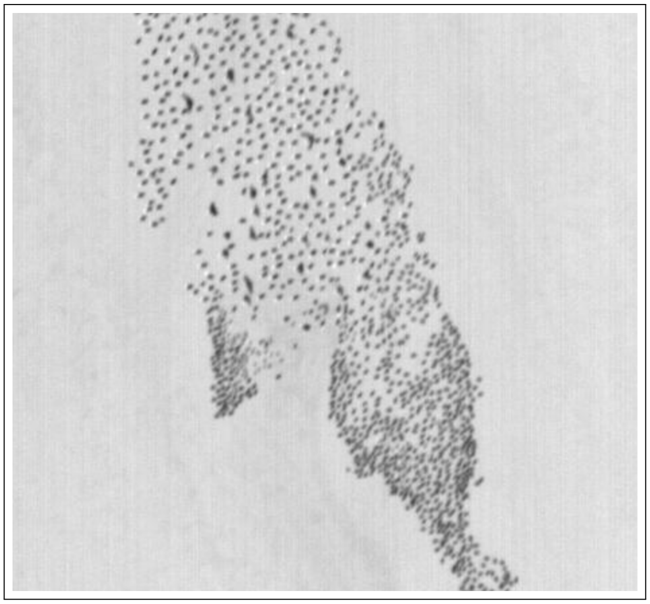


Figure 4: Example Tile of Mixed Image of Ducks/Geese and Sandhill Cranes

will be able to evaluate the confident of the model by generating multiple predictions on the same image. However, incorporating MC-Dropout into YOLO framework requires heavy modification of the open-source code. Thus, we will assess the feasibility of implementing MC-Dropout in YOLOv5 and YOLOv8. If not, we will be able to adopt Stochastic-YOLO, which is a MC-Dropout model on YOLOv3 [Azevedo et al., 2020].

Ensemble models are, by nature, computationally expensive. Since we are combining multiple individual models to make joint predictions, we are required to train multiple models individually. For instance, the architecture of YOLOv5x, which is the largest YOLOv5 model, consists of 86.7 million trainable parameters. Depending on the exact time required to train and run inference, ensemble modeling may or may not be a suitable approach.

Stratified K-Fold validation is a cross-validation technique used in classification to ensure each fold reflects the overall distribution of class labels. This method divides the dataset into K equal parts, maintaining the same proportion of each class as in the full dataset. The model is trained on $K - 1$ folds and tested on the remaining fold, rotating until each fold serves as a test set once. This approach is especially beneficial for handling imbalanced datasets, as it prevents bias in model evaluation by consistently representing all classes. Once every method implemented, we will compare the advantages and limitations of each and report results of the best ones. Below is a pseudocode for stratified K-Fold validation.

3.4 Assessing Model Generalizability

For this model to be applicable in the real world, we desire it to generalize into unseen future data. In other words, if we were to conduct the annual survey in the subsequent years (with similar conditions), we would want it to perform well on the newly acquired imageries. This will allows us to train only a small portion of new data without the need of repeating time-consuming process previously done.

To test model generalizability, we will use the 2018/2021 dataset as a benchmark comparison to the model fitted on 2023 dataset. Below is a brief table of configurations of model to assess generaliazbility.

Table 1: Training and Testing Configurations for Models

Model	Trained on	Tested on
YOLOv5x	2023 Dataset	2018 Dataset, 2021 Dataset
YOLOv8x	2023 Dataset	2018 Dataset, 2021 Dataset
RT-DETR	2023 Dataset	2018 Dataset, 2021 Dataset

Algorithm 1 Stratified k-Fold Validation

```
Input: Data D, Number of folds k
Output: Stratified partitions of the dataset
procedure STRATIFIEDKFOLD(D, k)
    Initialize folds to empty list of k lists
    Split D into classes C_1, C_2, ..., C_n
    for each class C_i do
        Shuffle C_i
        Divide C_i into k approximately equal parts
        for each part p do
            Assign p to a fold in folds
        end for
    end for
    Ensure each fold has approximately equal class proportions
    for each fold f in folds do
        train ← all data in folds except f
        test ← f
        Evaluate model on train and validate on test
    end for
end procedure
```

Some limitations of this experiment arises from the inherent dataset discrepancy across the years. For example, the survey methods of 2018 does not exactly align with 2021 or 2023. Also, images were taken at different altitude level which will definitely affect the model predictions. Hence, a careful consideration must be given on choosing which images to train and test on. Note that this only serves as a demonstration of model robustness. An interesting future research might be on collecting and modeling based on new survey imageries, comparing with the current results.

4 Prospective Project Timeline

The majority of this research will be completed during the Summer of 2024, with all final deliverables completed and turned in to USFWS and USGS by the end of the Fall, 2024 semester. Note that the project timeline may change depending on the progress and results achieved.

Below, I have outlined a basic timeline for the completion of major tasks:

Table 2: Project Timeline

Date	Activity	Status
January 2024	CRP 2024 Program Begins	
February-May 2024	Research objectives, literature review, dataset curation, initial modeling experiments	
June 2024	Error analysis, large-scale inference, finalizing dataset curation	
July 2024	Conduct uncertainty quantification following the methods described	
August 2024	Organize components to build a pipeline and finalize analysis of the results	
Fall 2024 Semester	Write a technical report or equivalent deliverables	
End of Fall 2024	Presentation of research and project wrap-up	

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